

## 20.5 Masking and swamping

Every diagnostic so far removes *one* case. But bad cases often come in groups: a batch of records in the wrong units, or a sensor that failed for an afternoon. **Masking** is when a bad case escapes detection because others hide it: deleting it alone leaves its companions to hold the fit in place. **Swamping** is when good cases are flagged because the bad ones have dragged the fit away from them. This section deletes sets of cases and proves how a cluster masks itself and swamps the rest.

### 20.5.1 Deleting a set of cases

Let  $D \subset \{1, \dots, n\}$  be a set of  $d$  cases. Write  $\mathbf{X}_D$  for the  $d \times p$  matrix of their rows,  $\mathbf{X}_{(D)}$  for  $\mathbf{X}$  with those rows removed,  $\hat{\mathbf{e}}_D$  for their residuals in the full fit, and

$$\mathbf{H}_D = \mathbf{X}_D(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}_D^\top$$

for the  $d \times d$  block of the hat matrix that belongs to them. Let  $\hat{\boldsymbol{\beta}}_{(D)}$  and  $\text{SSE}_{(D)}$  come from the least squares fit to the cases not in  $D$ , and let  $\mathbf{E}_D$  be the  $n \times d$  matrix whose columns are the coordinate vectors  $\mathbf{e}_i$ ,  $i \in D$ .

#### Theorem 20.5.1 — Deleting a set of cases

Let  $\mathbf{X}$  and  $\mathbf{X}_{(D)}$  have full column rank. Then  $\mathbf{I} - \mathbf{H}_D$  is nonsingular, and

- (a)  $\mathbf{y}_D - \mathbf{X}_D \hat{\boldsymbol{\beta}}_{(D)} = (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\mathbf{e}}_D$ ;
- (b)  $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(D)} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}_D^\top (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\mathbf{e}}_D$ ;
- (c)  $\text{SSE}_{(D)} = \text{SSE} - \hat{\mathbf{e}}_D^\top (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\mathbf{e}}_D$ ;
- (d) if  $n - p - d \geq 1$ , then under Eq. 7.3.1

$$F_D = \frac{\hat{\mathbf{e}}_D^\top (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\mathbf{e}}_D / d}{\text{SSE}_{(D)} / (n - p - d)} \sim F(d, n - p - d),$$

and  $F_D$  is the  $F$  statistic for  $\boldsymbol{\gamma} = \mathbf{0}$  in the mean-shift model  $\mathbf{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta} + \mathbf{E}_D\boldsymbol{\gamma}$ .

*Proof.* Let  $\mathbf{Z} = [\mathbf{X}, \mathbf{E}_D]$ . If  $\mathbf{X}\mathbf{b} + \mathbf{E}_D\mathbf{c} = \mathbf{0}$ , the rows outside  $D$  give  $\mathbf{X}_{(D)}\mathbf{b} = \mathbf{0}$ , so  $\mathbf{b} = \mathbf{0}$ , and then  $\mathbf{c} = \mathbf{0}$ . So  $\mathbf{Z}$  has full column rank. By Lemma 6.6.1(b) with  $\mathbf{X}_1 = \mathbf{E}_D$ , the matrix  $(\mathbf{I} - \mathbf{M})\mathbf{E}_D$  has full column rank, so its Gram matrix  $\mathbf{E}_D^\top (\mathbf{I} - \mathbf{M}) \mathbf{E}_D = \mathbf{I} - \mathbf{H}_D$  is nonsingular.

Fit the mean-shift model. By Theorem 6.6.2(a) with  $\mathbf{X}_1 = \mathbf{E}_D$  and  $\mathbf{X}_2 = \mathbf{X}$ , the estimated shifts are

$$\hat{\boldsymbol{\gamma}} = (\mathbf{E}_D^\top (\mathbf{I} - \mathbf{M}) \mathbf{E}_D)^{-1} \mathbf{E}_D^\top (\mathbf{I} - \mathbf{M}) \mathbf{y} = (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\mathbf{e}}_D.$$

On the other hand, for any  $\mathbf{b}$  the residual sum of squares  $\sum_{i \notin D} (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2 + \|\mathbf{y}_D - \mathbf{X}_D \mathbf{b} - \mathbf{c}\|^2$  is minimized over  $\mathbf{c}$  by  $\mathbf{c} = \mathbf{y}_D - \mathbf{X}_D \mathbf{b}$ , which removes the second term. What remains is minimized by  $\mathbf{b} = \hat{\boldsymbol{\beta}}_{(D)}$ , uniquely because  $\mathbf{X}_{(D)}$  has full column rank. So the mean-shift fit has coefficients  $\hat{\boldsymbol{\beta}}_{(D)}$  and  $\hat{\boldsymbol{\gamma}} = \mathbf{y}_D - \mathbf{X}_D \hat{\boldsymbol{\beta}}_{(D)}$ , and its residual sum of squares

is  $\text{SSE}_{(D)}$ . Comparing the two expressions for  $\hat{\mathbf{y}}$  gives (a). By Theorem 6.6.2(c), the coefficients of  $\mathbf{X}$  in the mean-shift fit are  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{y} - \mathbf{E}_D \hat{\mathbf{y}}) = \hat{\boldsymbol{\beta}} - (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}_D^\top \hat{\mathbf{y}}$ , which gives (b). By Lemma 6.6.1(c) the projection onto  $\mathcal{C}(\mathbf{Z})$  is  $\mathbf{M}$  plus the projection onto  $\mathcal{C}((\mathbf{I} - \mathbf{M})\mathbf{E}_D)$ , so the fall in the residual sum of squares from the fit of  $\mathbf{X}$  to the fit of  $\mathbf{Z}$  is

$$\mathbf{y}^\top (\mathbf{I} - \mathbf{M})\mathbf{E}_D(\mathbf{I} - \mathbf{H}_D)^{-1} \mathbf{E}_D^\top (\mathbf{I} - \mathbf{M})\mathbf{y} = \hat{\boldsymbol{\epsilon}}_D^\top (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\boldsymbol{\epsilon}}_D,$$

which gives (c). Finally  $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$  with dimensions  $p$  and  $p+d$ , and  $F_D$  is the statistic Eq. 11.1.1 for this pair of nested models, so (d) is Theorem 11.1.1(d). ■

With  $d = 1$  this is Theorem 20.3.1, and  $F_D = t_i^2$  (Exercise (A2)). Part (d) is exact only for a set chosen in advance; a Bonferroni correction over all  $\binom{n}{d}$  subsets is valid but very conservative.

## 20.5.2 How a cluster masks itself

The simplest mechanism is a group of identical bad cases:  $n_0$  clean cases plus  $m$  copies of one bad case.

### Proposition 20.5.2 — Masking and swamping by a cluster

Let  $\mathbf{X}_0$  ( $n_0 \times p$ , full column rank) and  $\mathbf{y}_0$  be the clean cases, with least squares estimate  $\hat{\boldsymbol{\beta}}_0$ , residuals  $\check{\epsilon}_j = y_j - \mathbf{x}_{(j)}^\top \hat{\boldsymbol{\beta}}_0$  and residual sum of squares  $\text{SSE}_0$ . Add  $m \geq 1$  cases, each with row  $\mathbf{x}_\star^\top$  and response  $y_\star$ . Put  $\mathbf{A} = \mathbf{X}_0^\top \mathbf{X}_0$ ,

$$h_\star = \mathbf{x}_\star^\top \mathbf{A}^{-1} \mathbf{x}_\star, \quad \delta = y_\star - \mathbf{x}_\star^\top \hat{\boldsymbol{\beta}}_0,$$

the leverage of  $\mathbf{x}_\star$  relative to the clean design and the distance of the cluster from the clean fit. In the least squares fit to all  $n_0 + m$  cases:

- (a) each cluster case has leverage  $h = h_\star / (1 + mh_\star) < 1/m$ ;
- (b) each cluster case has residual  $\delta / (1 + mh_\star)$ ;
- (c) deleting one cluster case gives it prediction residual  $\delta / (1 + (m-1)h_\star)$  and changes the estimate by  $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)} = \mathbf{A}^{-1} \mathbf{x}_\star \delta / [(1 + mh_\star)(1 + (m-1)h_\star)]$ ;
- (d) deleting the whole cluster gives each member prediction residual  $\delta$ , and  $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(D)} = m(1 + (m-1)h_\star)(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)})$ ;
- (e) clean case  $j$  has residual  $\check{\epsilon}_j - m\delta h_{j\star} / (1 + mh_\star)$ , where  $h_{j\star} = \mathbf{x}_{(j)}^\top \mathbf{A}^{-1} \mathbf{x}_\star$ , and  $\text{SSE} = \text{SSE}_0 + m\delta^2 / (1 + mh_\star)$ .

*Proof.* Let  $D$  be the cluster. Then  $\mathbf{X}^\top \mathbf{X} = \mathbf{A} + m\mathbf{x}_\star \mathbf{x}_\star^\top$ , and  $(\mathbf{A} + m\mathbf{x}_\star \mathbf{x}_\star^\top) \mathbf{A}^{-1} \mathbf{x}_\star = (1 + mh_\star) \mathbf{x}_\star$ , so

$$(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_\star = \frac{\mathbf{A}^{-1} \mathbf{x}_\star}{1 + mh_\star}. \quad (20.5.1)$$

- (a)  $h = \mathbf{x}_*^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_* = h_*/(1 + mh_*)$ , and  $mh = mh_*/(1 + mh_*) < 1$ .
- (b) and (d) Deleting  $D$  leaves the clean cases, so  $\hat{\boldsymbol{\beta}}_{(D)} = \hat{\boldsymbol{\beta}}_0$  and every cluster case has prediction residual  $\delta$ . All rows of  $\mathbf{X}_D$  equal  $\mathbf{x}_*^\top$ , so  $\mathbf{H}_D = h \mathbf{1} \mathbf{1}^\top$ , and by Theorem 20.5.1(a)  $\hat{\boldsymbol{\epsilon}}_D = (\mathbf{I} - h \mathbf{1} \mathbf{1}^\top) \delta \mathbf{1} = (1 - mh) \delta \mathbf{1} = \delta \mathbf{1}/(1 + mh_*)$ . By Theorem 20.5.1(b) and Eq. 20.5.1,  $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_0 = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}_D^\top \delta \mathbf{1} = m\delta(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_* = m\delta \mathbf{A}^{-1} \mathbf{x}_*/(1 + mh_*)$ .
- (c) By Theorem 20.3.1(d) and (b), with  $1 - h = (1 + (m - 1)h_*)/(1 + mh_*)$ , the prediction residual is  $\delta/(1 + (m - 1)h_*)$ . By Theorem 20.3.1(b) and Eq. 20.5.1,  $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_* \cdot \delta/(1 + (m - 1)h_*)$ , which is the stated vector. Comparing with the expression for  $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_0$  in the previous step gives the ratio in (d).
- (d)  $y_j - \mathbf{x}_{(j)}^\top \hat{\boldsymbol{\beta}} = \check{\epsilon}_j - \mathbf{x}_{(j)}^\top (\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_0)$ , and  $\mathbf{x}_{(j)}^\top (\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_0) = m\delta h_{j*}/(1 + mh_*)$ . For the sum of squares, Theorem 20.5.1(c) gives  $\text{SSE} = \text{SSE}_0 + \hat{\boldsymbol{\epsilon}}_D^\top (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\boldsymbol{\epsilon}}_D$ , and  $\hat{\boldsymbol{\epsilon}}_D^\top (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\boldsymbol{\epsilon}}_D = \delta^2 \mathbf{1}^\top (\mathbf{I} - h \mathbf{1} \mathbf{1}^\top)^{-1} \mathbf{1} = m\delta^2(1 - mh)$ , using  $(\mathbf{I} - h \mathbf{1} \mathbf{1}^\top)^{-1} \hat{\boldsymbol{\epsilon}}_D = \delta \mathbf{1}$  from part (b).

■

Each part is a way of hiding. **Leverage is masked** (a): however remote the cluster, each copy has leverage below  $1/m$ ; a single copy would have  $h_*/(1 + h_*)$ , near one. **Residuals are masked** (b): each is only  $1/(1 + mh_*)$  of the cluster's distance from the clean fit. **Deletion is masked** (c, d): deleting one copy leaves  $m - 1$  to hold the fit, recovering only  $1/(1 + (m - 1)h_*)$  of  $\delta$ . **Good cases are swamped** (e): clean residuals shift in proportion to their cross-leverage with the cluster, and  $s^2$  is inflated.

#### Example. (Four identical bad records)

Thirty clean cases have  $x$  uniform on  $(0, 10)$  and  $y = 2 + 0.5x + \varepsilon$ ,  $\sigma = 0.5$ . Four identical records are added at  $x = 20$ ,  $y = 2$ , where the clean line predicts about 12 (Figure 20.5.1(a)). The slope drops from 0.514 (clean cases) to 0.007, and  $s$  rises from 0.608 to 1.632. Here  $h_* = 1.126$  (it can exceed one, since  $x = 20$  is outside the clean design, as in Proposition 12.3.2) and  $\delta = -10.150$ . Each bad case has leverage 0.205, below  $1/m = 0.25$ , where a single copy would have 0.530; residual  $-1.844$ ; prediction residual  $-2.318$ ; and  $t_i = -1.28$ . Deleting the group changes  $\hat{\boldsymbol{\beta}}$  17.51 times as much as deleting one member.

The single-case diagnostics fail as predicted. The Bonferroni test (critical value 3.49) flags nothing, and the only case with  $|t_i| > 2$  is a clean one: case 30, the clean case with largest  $x$  (9.86), has  $t_i = 2.25$ , because the dragged line passes below it. Cook's distance does a little better: each bad case has  $D_i = 0.206$ , above  $4/n = 0.118$  (no clean case exceeds it; the largest is 0.100) but far below the median of  $F(2, 32)$ . A single bad record alone would have had  $t_i = -11.45$  and  $D_i = 13.46$ . Deleting the four together tells the truth: their prediction residuals are all  $\delta$ , and Theorem 20.5.1(d) gives  $F = 50.6$  on 4 and 28 degrees of freedom,  $p$ -value  $2.0 \times 10^{-12}$ . The difficulty is not in testing the cluster but in finding it.

```
import numpy as np
from scipy import stats

def single_case(X, y):
    """Leverages, residuals, s^2, externally studentized residuals and Cook's
```

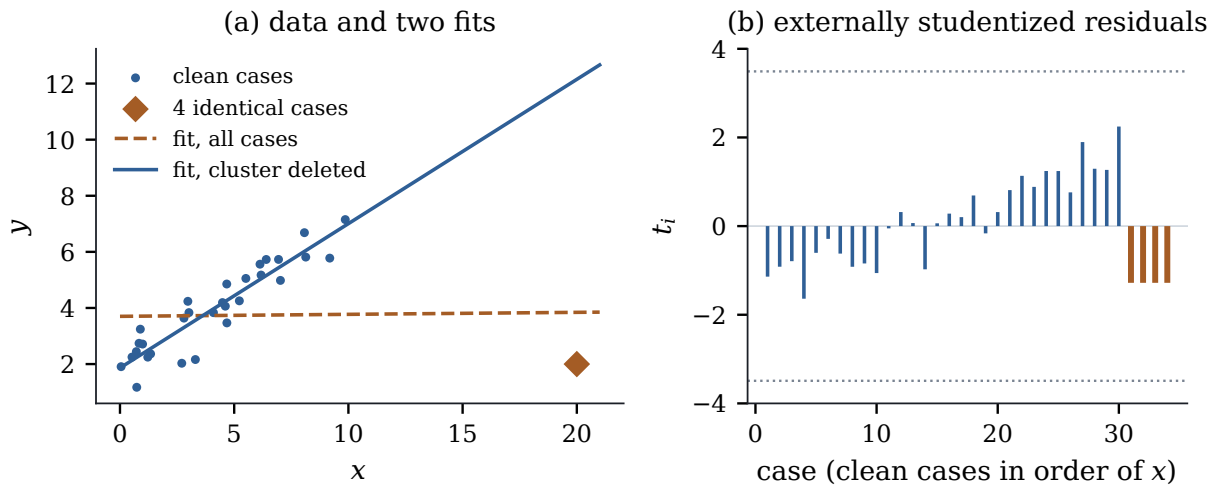


Figure 20.5.1: Masking and swamping. (a) Clean cases, four identical bad records (diamond), and least squares lines with and without them. (b) Externally studentized residuals (bad records last) and the 5% Bonferroni cut-offs.

```

distances. """
n, p = X.shape
Q, _ = np.linalg.qr(X)
h = np.sum(Q ** 2, axis=1)
e = y - Q @ (Q.T @ y)
s2 = e @ e / (n - p)
s2_del = (e @ e - e ** 2 / (1 - h)) / (n - p - 1)
t = e / np.sqrt(s2_del * (1 - h))
cook = e ** 2 * h / (p * s2 * (1 - h) ** 2)
return h, e, s2, t, cook

rng = np.random.default_rng(20)
n0, m = 30, 4                                # clean cases, size of the
cluster
x_clean = np.sort(rng.uniform(0, 10, n0))
y_clean = 2 + 0.5 * x_clean + rng.normal(0, 0.5, n0)
x0, y0 = 20.0, 2.0                            # four identical records
x = np.r_[x_clean, np.full(m, x0)]
y = np.r_[y_clean, np.full(m, y0)]
X = np.column_stack([np.ones(len(x)), x])
n, p = X.shape
cluster = np.arange(n0, n)

h, e, s2, t, cook = single_case(X, y)
crit = stats.t.isf(0.05 / (2 * n), n - p - 1)    # Bonferroni outlier test at 5%
print("slope with the cluster:", np.linalg.lstsq(X, y, rcond=None)[0][1])
print("cluster: h =", h[-1].round(3), " t =", t[-1].round(2), " Cook's D =", cook
      [-1].round(3))
i = int(np.argmax(np.abs(t)))
print(f"largest |t|: case {i + 1} at x = {x[i]:.2f}, t = {t[i]:.2f}; Bonferroni cut-
      off {crit:.2f}")

```

[Run this code online](#)

```

D = cluster                                # delete the whole cluster at
once
K = np.linalg.inv(X.T @ X)

```

```

H_D = X[D] @ K @ X[D].T
pred_D = np.linalg.solve(np.eye(m) - H_D, e[D])    # y_D minus its prediction from
the rest
sse = e @ e
sse_D = sse - e[D] @ pred_D
F = (e[D] @ pred_D / m) / (sse_D / (n - p - m))
print("group prediction residuals:", pred_D.round(3))
print(f"F = {F:.1f} on {m} and {n - p - m} degrees of freedom,"
      f" p-value {stats.f.sf(F, m, n - p - m):.1e}")

```

[Run this code online](#)

### 20.5.3 Finding the set

There are  $\binom{n}{d}$  candidate sets of each size  $d$ , far too many to search. The natural shortcut, deleting the case with the largest  $|t_i|$  and repeating, starts in the wrong place: in [Example \(Four identical bad records\)](#) it would first remove case 30, a clean case. Three ideas get around the combinatorics. First, **start from a fit the cluster cannot capture**, one determined by about half the data (the next section); the cases it flags form a candidate set to examine and test, remembering that it was chosen by looking. Second, **grow a clean subset**: the forward search of Atkinson and Riani (2000) adds cases one at a time in order of their residuals from a robust start, and a cluster enters late and all at once, as a jump in the monitored residuals and estimates; Hadi and Simonoff (1993) proposed a related procedure. Third, **unmask leverage separately**: by [Proposition 20.5.2\(a\)](#) the hat matrix is fooled by clusters, since leverage is a Mahalanobis distance built from the sample covariance, which a cluster inflates in its own direction. Rousseeuw and van Zomeren (1990) replaced the mean and covariance by estimates that ignore up to half the cases.

*Idea.* Single-case diagnostics answer “what if this case alone were different?”, the wrong question when bad cases come in groups. Group deletion asks the right one, but only for a suspected group; finding the group needs a fit it cannot capture.

### 20.5.4 Exercises

#### 20.5.4.1 A. Check your understanding

##### Exercise A1.

Check the leverage 0.205 and the residual  $-1.844$  in [Example \(Four identical bad records\)](#) from  $h_\star = 1.126$ ,  $\delta = -10.150$  and  $m = 4$ .

##### Exercise A2.

Show that [Theorem 20.5.1](#) with  $D = \{i\}$  gives parts (b), (d) and (e) of [Theorem 20.3.1](#), and that  $F_D = t_i^2$ .

*Solution.* With  $d = 1$ ,  $\mathbf{H}_D = h_{ii}$  and  $\hat{\mathbf{e}}_D = \hat{\varepsilon}_i$ , so (a) is the prediction residual  $\hat{\varepsilon}_i/(1 - h_{ii})$ , (b) is [Theorem 20.3.1\(b\)](#), and (c) is  $\text{SSE}_{(i)} = \text{SSE} - \hat{\varepsilon}_i^2/(1 - h_{ii})$ . Then  $F_D = \hat{\varepsilon}_i^2/\{(1 -$

$$\{h_{ii}s_{(i)}^2\} = t_i^2.$$

### 20.5.4.2 B. Practice

#### Exercise B1.

Show that for  $\mathbf{H}_D = h\mathbf{1}\mathbf{1}^\top$  ( $d \times d$ ) with  $dh \neq 1$ ,  $(\mathbf{I} - h\mathbf{1}\mathbf{1}^\top)^{-1} = \mathbf{I} + h\mathbf{1}\mathbf{1}^\top/(1 - dh)$ , and use it to recompute the group term  $\hat{\mathbf{e}}_D^\top(\mathbf{I} - \mathbf{H}_D)^{-1}\hat{\mathbf{e}}_D$  in the proof of [Proposition 20.5.2\(e\)](#) from  $\hat{\mathbf{e}}_D = \delta\mathbf{1}/(1 + mh_*)$ .

#### Exercise B2.

In simple regression with clean mean  $\bar{x}_0$  and  $S_0 = \sum(x_j - \bar{x}_0)^2$ , show that  $h_{j*} = 1/n_0 + (x_j - \bar{x}_0)(x_* - \bar{x}_0)/S_0$ . In [Example \(Four identical bad records\)](#), with  $x_*$  far to the right and  $\delta < 0$ , which clean residuals does [Proposition 20.5.2\(e\)](#) push up, and which down? Explain why case 30 was flagged.

*Solution.* Use the centred rows  $(1, x - \bar{x}_0)$ , which span the same column space; then  $\mathbf{A}^{-1} = \text{diag}(1/n_0, 1/S_0)$ , so  $h_{j*} = 1/n_0 + (x_j - \bar{x}_0)(x_* - \bar{x}_0)/S_0$ . The shift in clean residual  $j$  is  $-m\delta h_{j*}/(1 + mh_*)$ , which has the sign of  $h_{j*}$  because  $\delta < 0$ . Since  $x_* - \bar{x}_0 > 0$ ,  $h_{j*}$  grows with  $x_j$ : positive on the right of the clean data and negative on the far left (once  $(x_j - \bar{x}_0)(x_* - \bar{x}_0)/S_0 < -1/n_0$ ). Residuals on the right are pushed up and those on the far left down. Case 30 is the clean case with the largest  $x$ , so it has the largest upward shift, and its studentized residual crosses two.

#### Exercise B3.

In the setting of [Proposition 20.5.2](#), delete  $k < m$  of the cluster cases. Show that each deleted case has prediction residual  $\delta/(1 + (m - k)h_*)$ . What does this say about searching for a cluster of unknown size by deleting sets of size  $k$ ?

### 20.5.4.3 C. Going deeper

#### Exercise C1.

Place two clusters of  $m$  identical bad cases at the two ends of a straight-line design, one above the clean line and one below. Describe, using [Theorem 20.5.1](#), what single-case deletion, deletion of one cluster, and deletion of both clusters report. Is it possible for each cluster to mask the other?

#### Exercise C2.

For a set  $D$  of cases, define  $D_D = (\hat{\boldsymbol{\beta}}_{(D)} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\hat{\boldsymbol{\beta}}_{(D)} - \hat{\boldsymbol{\beta}})/(ps^2)$ . Using [Theorem 20.5.1\(b\)](#), show that  $D_D = \hat{\mathbf{e}}_D^\top(\mathbf{I} - \mathbf{H}_D)^{-1}\mathbf{H}_D(\mathbf{I} - \mathbf{H}_D)^{-1}\hat{\mathbf{e}}_D/(ps^2)$ .